

Vulnerability Scanner (Mini Project)

A simple Python script that detects common vulnerabilities on a host or web application.

What it does

- Scans common ports (FTP, SSH, HTTP, HTTPS, MySQL, RDP, etc.) to find open ones
- Grabs service banners and flags outdated software versions (OpenSSH, Apache, nginx, etc.)
- Flags weak configurations: risky exposed services and missing web security headers
- Generates a simple vulnerability report (`vuln_report.txt`)

How to run

Requires Python 3.8+ (no extra installs).

```
python vulnerability_scanner.py --demo
# safe local demo
python vulnerability_scanner.py 127.0.0.1
# scan a host
```

```
python vulnerability_scanner.py  
http://localhost:8000 # scan a web app
```

Sample output

See `vuln_report.txt` and `demo_output.txt` .

Ethical use

Scan only systems you own or have written permission to test.

Learning outcome

Basics of penetration testing and vulnerability assessment.